E. Boehm. Lecture on 709 etc April 8, 1958

(1) 701 – great deal of "button pushing" – has gone down continuously
– requirement for "halt" type
– Scaling prob

704 – housekeeping + floating point – subroutine technique.

709 – data access improved – indirect addressing.
– "exente" feature
– Scatter read + write in 701-704 (simultaneous read/write)
– Data Streaming Unit – { given flexible control to do Scatter Read/write, control word
Interrupt feature – I/O interrupts -- I/O does not interrupt now.

Difference in Tape record controls program not just data.

Program compatibility: I/O area + minor floppoint changes.
– using a program + 1 machine change.

drums on 709 are not being ordered – 32K mem. is instead.

– as many as 48 tapes (in BCD) conversion flexible system
"convert" instruction. any radix to any other.
also useful for analyzing data in "streaming".
– limitation: – one wants more flexibility – can't stop itself + proceed.

eg. [SXA. instr. changes.
complements in mult+ij

SHARE org: prog. cost = cost of machine. standard RPQ's
eg. overflow – underflow in Fl point "trap" now on 709.
→ compatibility more important.
can't justify all details in advance – but become imp. later. (over)

operator type operation — less button setting.

36 Sense Indicators: Set from mem, acc
Store " "

Test or set in 18 bit groups

"blocking out" memory.

multiprogramming 709 not really fast enough.

I/O Record control.

Scatter read/write: control wds.

IOCT ← Sig. point — holds up + gets new wd.
IOCP

not Transmitting data to mem — for passing tape.

up to 6 channels can operate at once

← more internal checking — programs has built in checks —

commercial users about 8 or 9 now.

— G.E. Salem more complete calcs. in one run.

Meeting with E. Bloch, E. Foss, R. Merwin
. D. Sweeney — H. Nicholas
unified SIGMA approach.

April 8, '58

Present Design

(1) 3 in 1 concept instructions

(2) 3 in 1 concept hardware

(3) Shortcomings

(a) I-Box not optomized for I perf.

(b) current switching ckts only in Base

(c) Extensive VFL

(d) Transistor count 209,000.

New Org,

optimize System

(a) Eliminate E-I interconnection

(b) Modify Resister Org. ?

(c) Cut down stages in pipeline by virtue of EF logic

(d) EF logic & Cascode logic inherent advisable,

(e) Limit Mem fetches to indexing, limit and LA

(f) exp. adder used for VFL 12 bit

(g) centeral instr decoder (E box + AU)

others

- Simplify instrs. set ?
- progressive indexing ?
- decimal arith ?

results:

reduce transistor count

Foss — saving ~ 5000, Transistors at outside
Bloch — 3% approx

get I-Box perf. up to 0.8 μs rate

1 register = 2,000. transistors.

Transistor Count

Present		New	
31,000	I	28,000.	} reduced by console logic.
50,000 (44,000)	E	45,000. (39,000)	— E not incl consoles, etc.
2,000	I-R	2,000.	
38,000	LA	34,000.	
15,000	TR-IR	---	
35,000	AU (min	35,000?	
21,000	MPYR	21,000. (?)	2 μs Mpy vs 6.5 μs without
192,000		162,000.	marginal gates another 12,000. ??

Transistor Count × regs →

2 Machine philosophy. (Hard to get more in performance)

70XR Σ

- Possibility Base ~ 50,000, trs. if Sigma considerations were auth.

Speed	Machine	Rent	Sales
1	709	130	–
4	Transac	28	–
13	Larc(6.5M)	21	–
16	B (1M)	–	0
65	Σ	–	4

← Marketing Estimates